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(57) **ABSTRACT**

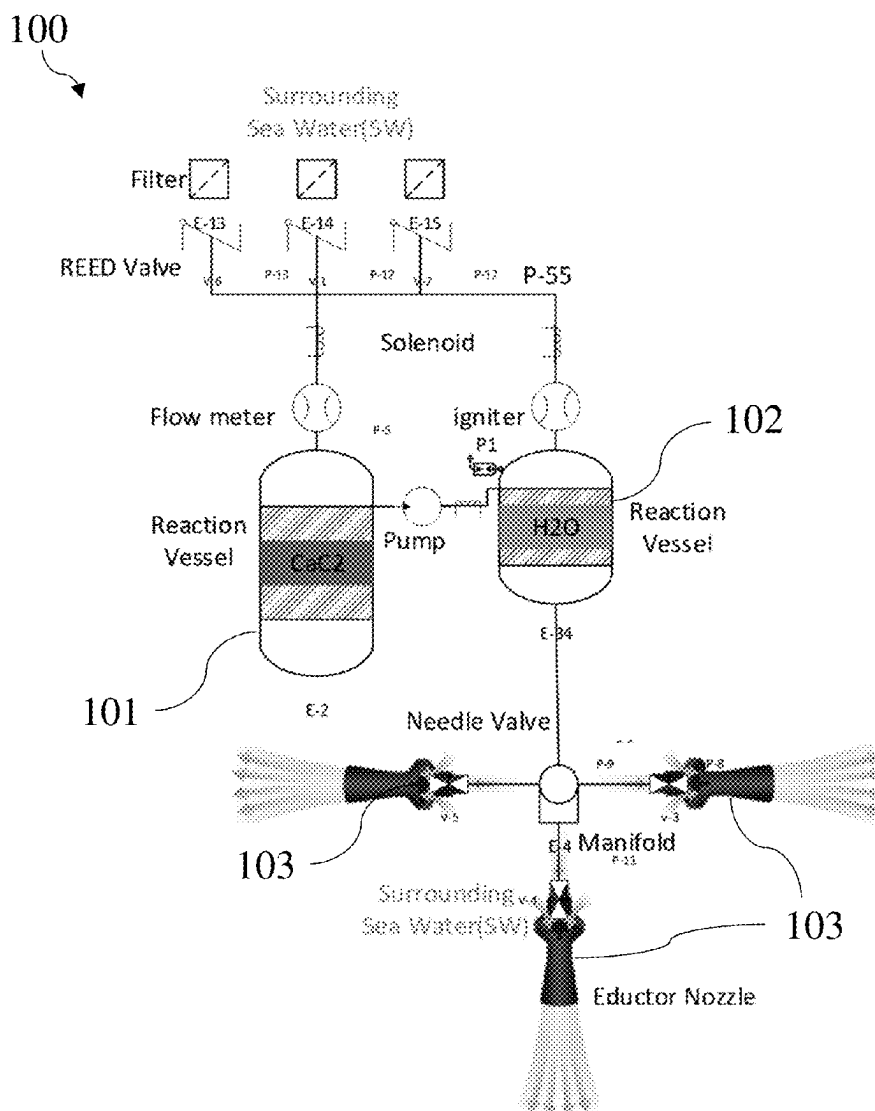
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Related U.S. Application Data

(60) Provisional application No. 63/560,192, filed on Mar. 1, 2024.

A method for underwater propulsion including a thermal engine system wherein the thermal engine system produces a propulsion gas, the propulsion gas is introduced into a pressure tuned valve, wherein the pressure tuned valve lies upstream of an eductor and the pressure tuned valve controls release of the gas from the thermal engine system to the eductor; and wherein the release of the propulsion gas to the eductor provides underwater propulsion by ejection of the propulsion gas into surrounding water.



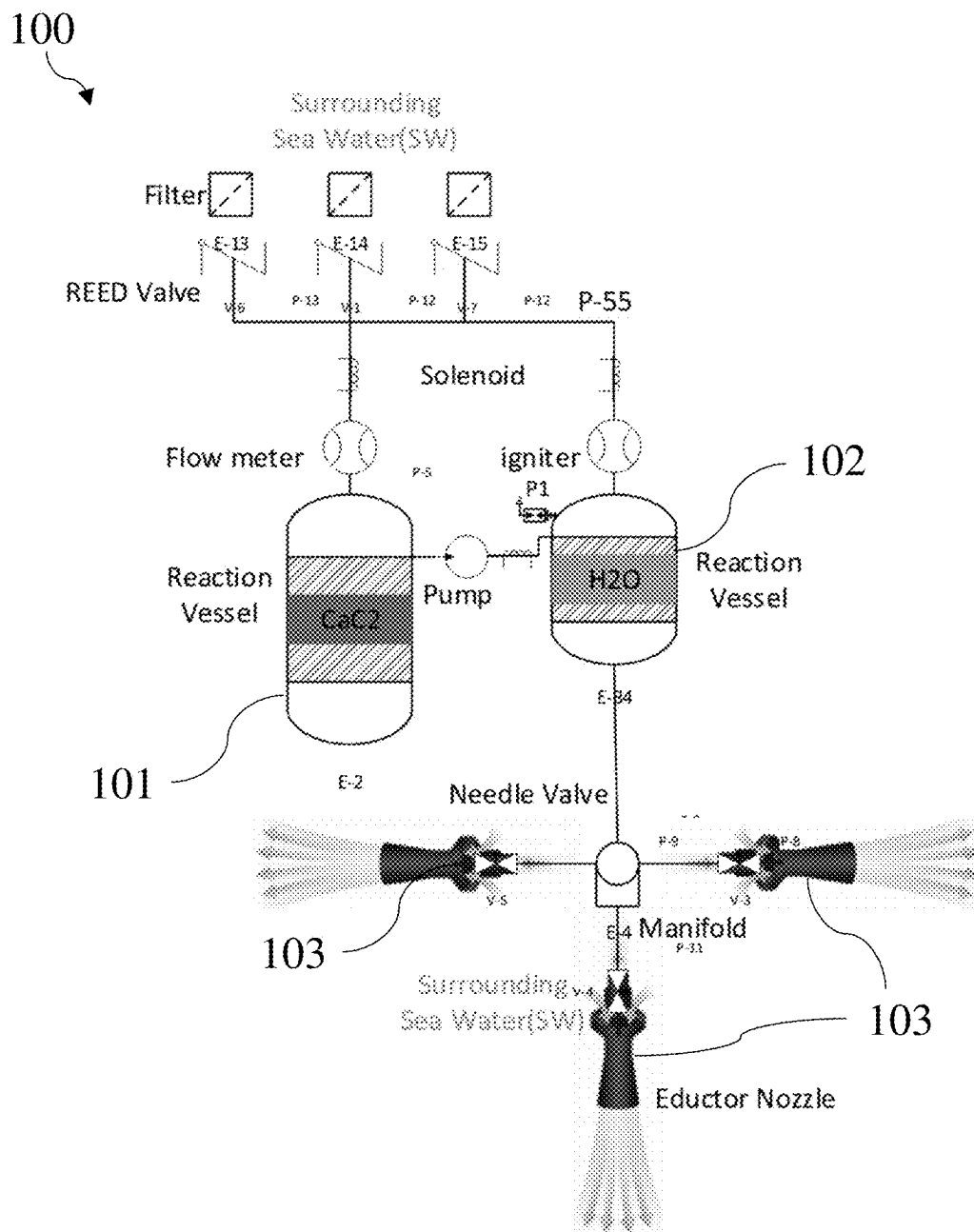


FIG. 1

200

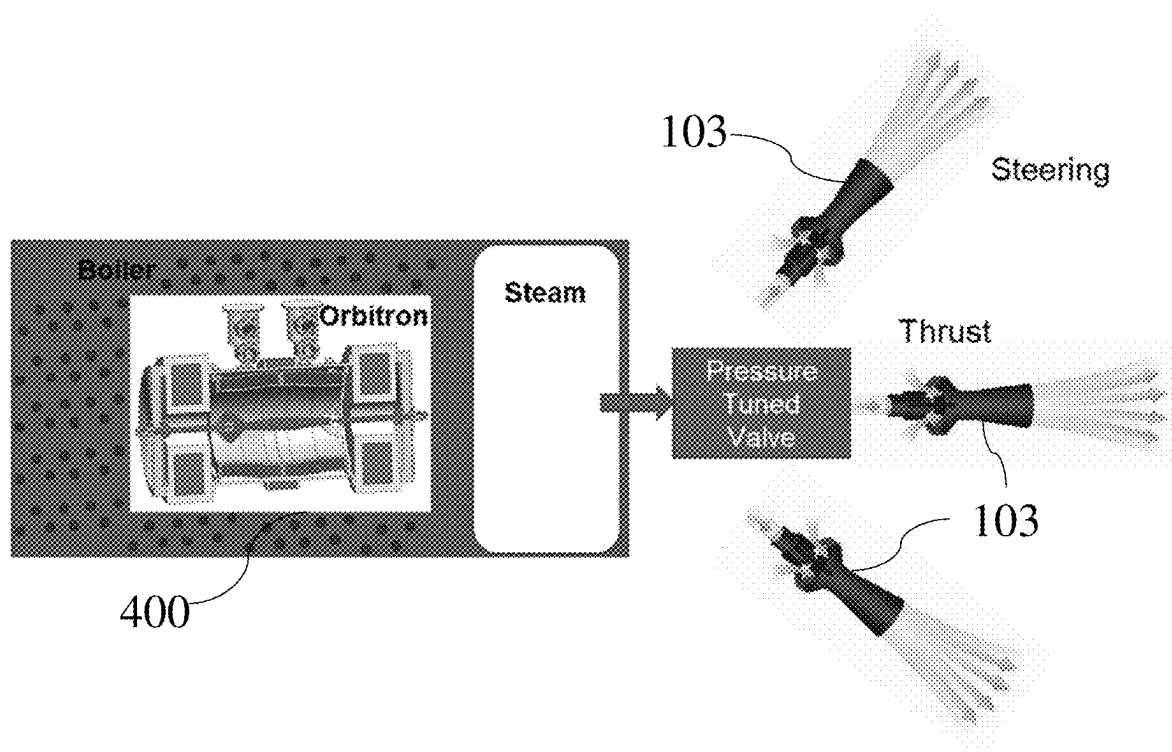


FIG. 2

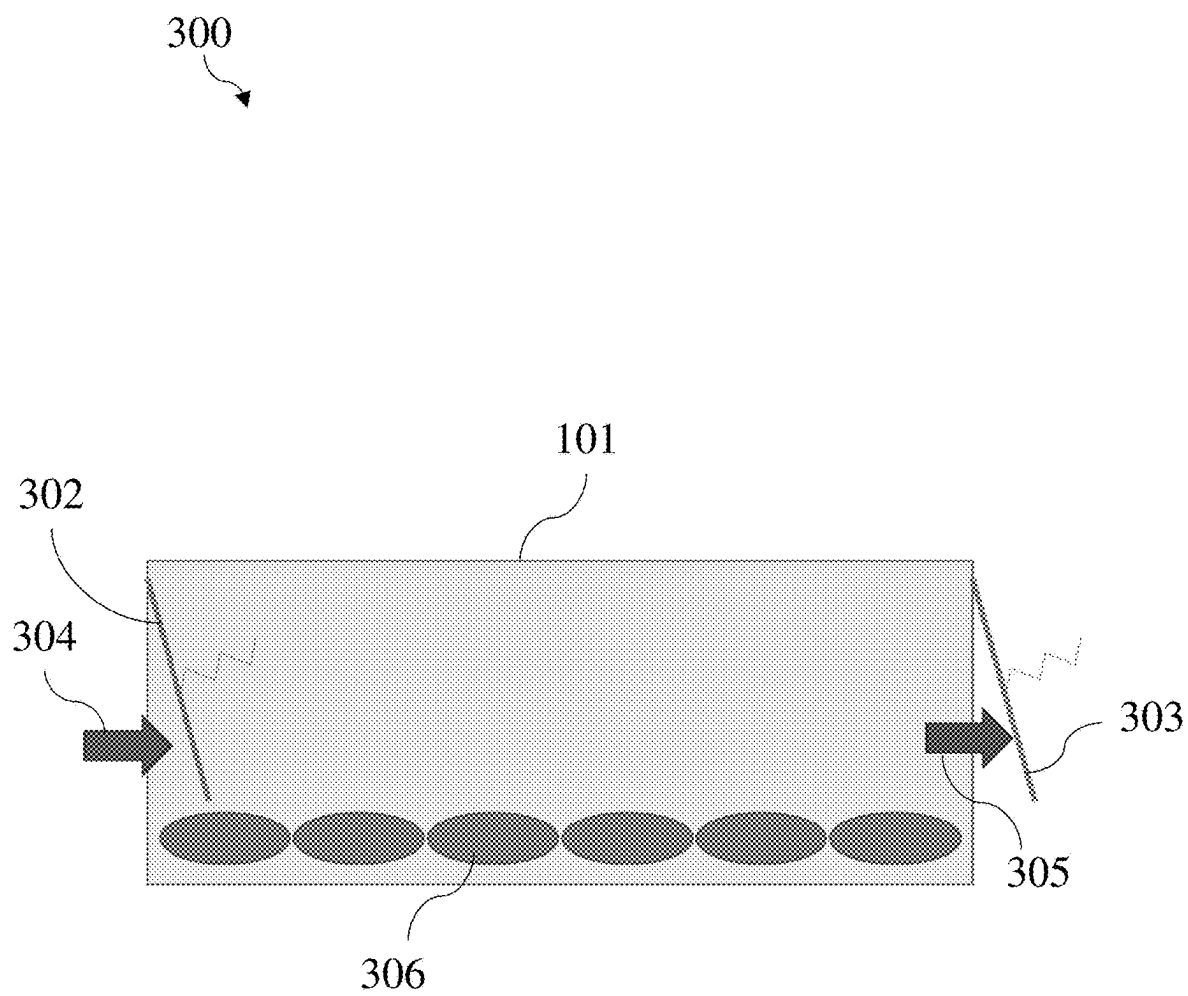


FIG. 3

400

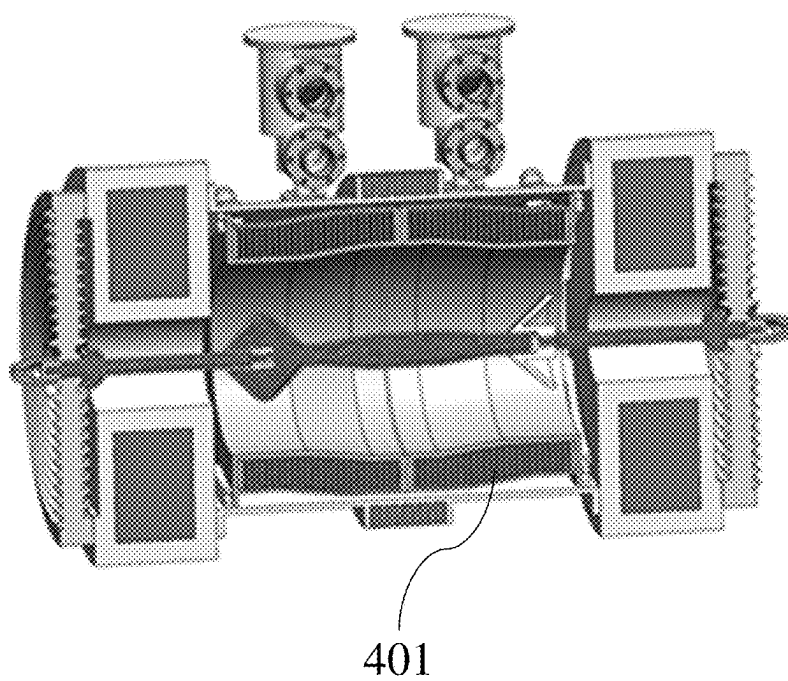



FIG. 4

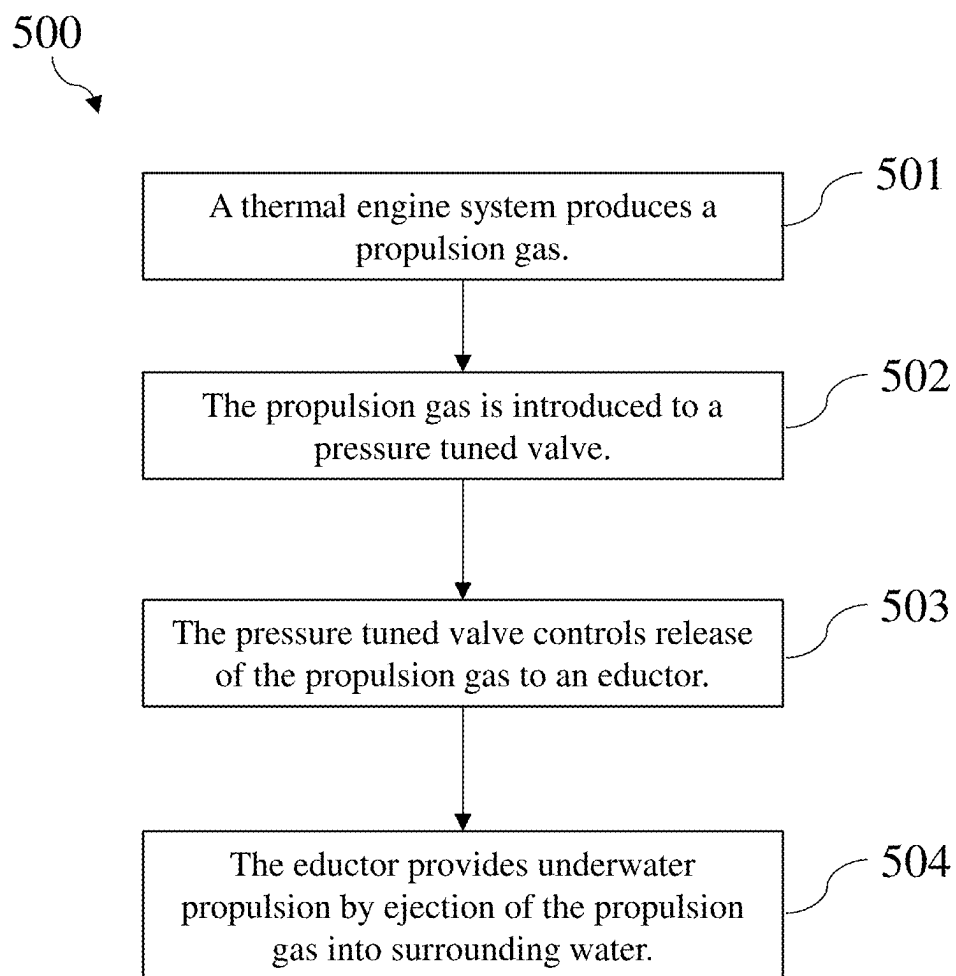


FIG. 5

THERMAL ENGINE FOR UNDERWATER PROPULSION AND METHODS OF USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/560,192 filed Mar. 1, 2024, which is incorporated by reference herein in its entirety.

GOVERNMENT LICENSE RIGHTS

[0002] This invention was made with Government support under Contract No. HR0011-23-9-0046 awarded by the United States Department of Defense. The Government has certain rights in the invention.

BACKGROUND

[0003] The subject matter disclosed herein generally relates to a thermal engine system, and methods of use thereof.

[0004] Adequate energy density for unmanned undersea vehicles (UUV) remains a challenge for efficient underwater propulsion. Existing system such as stored chemical energy propulsion systems (SCEPS) continue to have limited efficiency and inadequate speed propulsion.

[0005] A need remains for an undersea propulsion system that provides energy generation and for variable speed propulsion without loss of efficiency.

SUMMARY

[0006] A thermal engine system for underwater propulsion includes a first reaction chamber, a second reaction chamber, a pump, and an ignitor. The pump is downstream of the first reaction chamber and upstream of the second reaction chamber. The pump is in fluid communication with the first reaction chamber and the second reaction chamber and is configured to pump a first gas from the first reaction chamber to the second reaction chamber. The ignitor is operative to ignite the first gas to produce a second gas in the second reaction chamber. The second gas is in fluid communication with a pressure tuned valve. The pressure tuned valve lies upstream of an eductor and the pressure tuned valve is operative to release the second gas to the eductor. The eductor is configured to provide undersea propulsion by ejection of the steam into surrounding water.

[0007] A thermal engine system for underwater propulsion including a reaction chamber that is operative to generate heat from electrostatic fusion, wherein the reaction chamber includes a heat exchanger; a boiler surrounding the reaction chamber, wherein the boiler contains water; wherein the heat exchanger is operative to vaporize water into steam within the boiler with the energy generated from electrostatic fusion; a pressure tuned valve, wherein the steam is in fluid communication with the pressure tuned valve and the pressure tuned valve is operative to control the release of steam from the boiler. The pressure tuned valve lies downstream of the boiler and upstream of an eductor and the pressure tuned valve is operative to release the steam to the eductor. The eductor is configured to provide undersea propulsion by ejection of the steam into surrounding water.

[0008] A method for underwater propulsion with a thermal engine system that produces a propulsion gas, is disclosed. The method includes introducing the propulsion gas into a

pressure tuned valve. The pressure tuned valve lies upstream of an eductor and the pressure tuned valve controls release of the propulsion gas from the thermal engine system to the eductor. The release of the propulsion gas to the eductor provides underwater propulsion by ejection of the propulsion gas into surrounding water.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] The subject matter is particularly pointed out and distinctly claimed at the conclusion of the specification. The foregoing and other features, and advantages of the present disclosure are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

[0010] FIG. 1 is a schematic diagram of an embodiment of a thermal engine system;

[0011] FIG. 2 is a schematic diagram of an embodiment of a thermal engine system;

[0012] FIG. 3 is a cross sectional view of an embodiment of a chemical reaction chamber;

[0013] FIG. 4 is a cross sectional view of an embodiment of a microfusion reaction chamber; and

[0014] FIG. 5 is a flow diagram of a method for underwater propulsion including a thermal engine system.

DETAILED DESCRIPTION

[0015] The exemplary embodiments disclosed herein are illustrative of thermal engine systems for underwater propulsion and methods of use thereof. It should be understood, however, that the disclosed embodiments are merely examples of the present disclosure, which may be embodied in various forms. Therefore, details disclosed herein with reference to example abradable coatings, components coated with an abradable coating, and associated methods of use and manufacture thereof are not to be interpreted as limiting, but merely as the basis for teaching one skilled in the art about the abradable coatings and how to make and use the abradable coatings of the present disclosure.

[0016] Disclosed herein is a thermal engine system for underwater propulsion. The thermal engine system can be powered by a chemical, water reactive process or by an electrostatic fusion reaction. The processes for powering the system can be used to generate energy for propulsion. The energy can be used to convert water to steam or to generate carbon dioxide. The produced gas can be used for underwater propulsion via eductor thrust. The systems and methods disclosed herein provide higher energy density, lower costs, improved throttle, and so forth. Rather than turning of a turbine or ejection of water for propulsion, the systems and methods disclosed provide underwater propulsion via the ejection of small amounts of gas from an eductor. In some embodiments, the system further combines fusion energy to generate thermal energy for underwater propulsion.

[0017] FIG. 1 is a schematic diagram of an embodiment of a thermal engine system 100. As shown in FIG. 1, a reaction vessel (herein after a first reaction chamber 101), contains a chemical reactant (e.g., calcium carbide), the system can facilitate the flow of surrounding water [e.g., surrounding seawater (SW)] into the thermal engine system via reed valves as shown or via suitable control mechanisms. Flow can be monitored in the system via a flow meter. Water can be permitted to flow into the reaction chamber 101. The

chemical reactant and the water can react to form a first gas. The pump can be configured to pump the first gas from the first reaction chamber **101** to a second reaction chamber **102**. In the second reaction chamber **102**, the first gas can be ignited by an ignitor to produce the propulsion gas as a combustion byproduct. The system can have an oxygen-containing gas inlet into the second reaction chamber to facilitate the combustion of the first gas. The propulsion gas can be released to a pressure tuned valve downstream of the second reaction chamber. The pressure tuned valve is upstream of an eductor **103** and the pressure tuned valve can release the propulsion gas to the eductor **103**. In some embodiments, the propulsion gas can be released to the pressure tuned valve via a manifold as shown in FIG. **1**. The eductor can be configured to release the steam for underwater propulsion.

[0018] FIG. **2** is a schematic diagram of an embodiment of a thermal engine system **200**. As shown in FIG. **2**, a reaction chamber can contain a microfusion reactor **400**. The microfusion reactor can include a heat exchanger. The heat exchanger can be in fluid contact with water in a surrounding boiler. The heat exchanger can be configured to transfer the heat generated from electrostatic fusion to the water in the surrounding boiler to convert water in the boiler to steam. The steam (i.e., the propulsion gas) is in fluid contact with a pressure tuned valve downstream of the reaction chamber **101**. The pressure tuned valve is upstream of an eductor and in fluid contact with the eductor **103**. The pressure tuned valve can be configured to release the steam to the eductor. The eductor can be configured to release the steam for underwater propulsion.

[0019] FIG. **3** is a cross sectional view of an embodiment of a chemical reaction chamber **300**. In the chemical reaction chamber, the chemical reactant **306** is contained in the reaction chamber **101**. An inlet valve **302** (also called “a first valve” herein) can facilitate the flow of water through a first inlet into the reaction chamber **101**. An outlet valve **303** (also called “a second valve” herein) can facilitate the flow of the first gas out of a first outlet in the reaction chamber **101**. The flow of water into the reaction chamber is shown by arrow **304** and the flow of the first gas out of the reaction chamber is shown by arrow **305**. In some embodiments, the valve **302** and valve **303** can be reed valves. In operation, as the reed valve **302** opens to permit the flow of water into the reaction chamber **101**, the reed valve **303** opens to equalize the pressure. Upon pressure equalization, the valves close and the reaction can be permitted to proceed. The reed valve **303** can be set to a desired ejection pressure to open and release the first gas when the pressure is met.

[0020] The chemical reactant can include calcium carbide, lithium metal, sodium metal, potassium metal, or a combination thereof. The first gas can include acetylene, hydrogen, or a combination thereof. The second gas can include steam, carbon dioxide, or a mixture thereof.

[0021] FIG. **4** is a cross sectional view of an embodiment of a microfusion reaction chamber **400**. The microfusion reaction chamber **400** produces thermal energy. The microfusion reaction chamber includes a heat exchanger **401** to facilitate the removal of heat from the microfusion reaction chamber. The microfusion reaction chamber can generate fusion ion energy via an electrostatic field. In operation, the electrostatic field can trap fusion fuel ions in elliptical orbits, electrons are introduced into the reactor and confined via a

magnetron electron process, and nuclear fusion occurs when orbiting ions cross paths generating heat.

[0022] The thermal engine can include a microturbine generator to provide propulsion or generate energy from the flow of the propulsion gas.

[0023] FIG. **5** provides a flow diagram of a method **500** for underwater propulsion including a thermal engine system. The method **500** includes the production of a propulsion gas by a thermal engine system in step **501**. The propulsion gas is introduced to a pressure tuned valve (step **502**). The pressure tuned valve is upstream of an eductor. The pressure tuned valve controls the release of the propulsion gas to the eductor (step **503**). The eductor then produces underwater propulsion by ejection of the propulsion gas into surrounding water (step **504**).

[0024] In an embodiment, the method **500** can further include a thermal engine system with a first reaction chamber, a pump, a second reaction chamber, and an ignitor. A first gas can be generated by a chemical reaction between water and a chemical reactant in the first reaction chamber, the first gas can be pumped to the second reaction chamber to be ignited by the ignitor to produce the propulsion gas.

[0025] In another embodiment, the method **500** can further include a reaction chamber with a heat exchanger and a boiler surrounding the reaction chamber. The boiler can contain water in fluid contact with the reaction chamber. The reaction chamber can generate heat from electrostatic fusion, the heat exchanger can then transfer the heat from electrostatic fusion to the water in the boiler. The heat transfer can serve to vaporize the water into steam. The steam can be used as the propulsion gas.

[0026] The thermal engine system can further comprise a microturbine generator. The microturbine generator can generate power from the flow of the propulsion gas to the eductor.

[0027] The propulsion gas can be steam, carbon dioxide, or a combination thereof. The eductor can provide steering control, thrust, or a combination thereof. In some embodiments, the thermal engine system can include a plurality of eductors. The plurality of eductors can include 1 to 20 eductors, 3 to 5 eductors, and so forth. The plurality of eductors can provide steering, thrust, or a combination thereof.

[0028] While particular embodiments have been described, alternatives, modifications, variations, improvements, and substantial equivalents that are or may be presently unforeseen may arise to applicants or others skilled in the art. Accordingly, the appended claims as filed and as they may be amended are intended to embrace all such alternatives, modifications variations, improvements, and substantial equivalents.

[0029] The ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other (e.g., ranges of “up to 25 wt. %,” or, more specifically, 5 wt. % to 20 wt. %”, is inclusive of the endpoints and all intermediate values of the ranges of “5 wt. % to 25 wt. %,” and so forth). “Combinations” is inclusive of blends, mixtures, alloys, reaction products, and the like. The terms “first,” “second,” and the like, do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms “a” and “an” and “the” do not denote a limitation of quantity and are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by con-

text. “Or” means “and/or” unless clearly stated otherwise. Reference throughout the specification to “some embodiments”, “an embodiment”, and so forth, means that a particular element described in connection with the embodiment is included in at least one embodiment described herein, and may or may not be present in other embodiments. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various embodiments. A “combination thereof” is open and includes any combination comprising at least one of the listed components or properties optionally together with a like or equivalent component or property not listed.

[0030] Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this application belongs. All cited patents, patent applications, and other references are incorporated herein by reference in their entirety. However, if a term in the present application contradicts or conflicts with a term in the incorporated reference, the term from the present application takes precedence over the conflicting term from the incorporated reference.

[0031] Although the thermal engine systems and methods of the present disclosure have been described with reference to example embodiments thereof, the present disclosure is not limited to such example embodiments and/or implementations. Rather, the thermal engine systems and methods of the present disclosure are susceptible to many implementations and applications, as will be readily apparent to persons skilled in the art from the disclosure hereof. The present disclosure expressly encompasses such modifications, enhancements and/or variations of the disclosed embodiments. Since many changes could be made in the above construction and many widely different embodiments of this disclosure could be made without departing from the scope thereof, it is intended that all matter contained in the drawings and specification shall be interpreted as illustrative and not in a limiting sense. Additional modifications, changes, and substitutions are intended in the foregoing disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the disclosure.

What is claimed is:

1. A thermal engine system for underwater propulsion comprising

- a first reaction chamber;
- a second reaction chamber;
- a pump downstream of the first reaction chamber and upstream of the second reaction chamber, wherein the pump is in fluid communication with the first reaction chamber and the second reaction chamber and the pump is configured to pump a first gas from the first reaction chamber to the second reaction chamber;
- an ignitor that is operative to ignite the first gas to produce a second gas in the second reaction chamber;
- wherein the second gas is in fluid communication with a pressure tuned valve;
- wherein the pressure tuned valve lies upstream of an eductor and wherein the pressure tuned valve is operative to release the second gas to the eductor; and
- wherein the eductor is configured to provide undersea propulsion by ejection of the steam into surrounding water.

2. The thermal engine system of claim 1, further comprising a microturbine generator.

3. The thermal engine system of claim 1, wherein the first reaction chamber comprises a first inlet and a first outlet, wherein a first valve is configured to release water into the first reaction chamber through the first inlet and a second valve is configured to release a first gas from the first reaction chamber through the first outlet.

4. The thermal engine system of claim 1, wherein the first reaction chamber is configured to contain a chemical reactant that when combined with water undergoes a chemical reaction to generate the first gas.

5. The thermal engine system of claim 1, wherein the reaction chamber is configured to contain a chemical reactant, wherein the chemical reactant comprises calcium carbide, lithium metal, sodium metal, potassium metal, or a combination thereof, and wherein the first gas is acetylene, hydrogen, or a combination thereof.

6. The thermal engine system of claim 1, wherein the second gas is a mixture of steam and carbon dioxide.

7. A thermal engine system for underwater propulsion comprising

- a reaction chamber that is operative to generate heat from electrostatic fusion, wherein the reaction chamber comprises a heat exchanger;

- a boiler surrounding the reaction chamber, wherein the boiler contains water;

wherein the heat exchanger is operative to vaporize water into steam within the boiler with the energy generated from electrostatic fusion;

- a pressure tuned valve, wherein the steam is in fluid communication with the pressure tuned valve and the pressure tuned valve is operative to control the release of steam from the boiler;

wherein the pressure tuned valve lies downstream of the boiler and upstream of an eductor and wherein the pressure tuned valve is operative to release the steam to the eductor; and

wherein the eductor is configured to provide undersea propulsion by ejection of the steam into surrounding water.

8. The thermal engine system of claim 7, further comprising a microturbine generator.

9. A method for underwater propulsion comprising a thermal engine system wherein the thermal engine system produces a propulsion gas,

the propulsion gas is introduced into a pressure tuned valve, wherein the pressure tuned valve lies upstream of an eductor and the pressure tuned valve controls release of the gas from the thermal engine system to the eductor; and

wherein the release of the propulsion gas to the eductor provides underwater propulsion by ejection of the propulsion gas into surrounding water.

10. The method of claim 9, wherein the thermal engine system further comprises a first reaction chamber; a pump; a second reaction chamber; and an ignitor;

wherein a first gas is generated by a chemical reaction between water and a chemical reactant in the first reaction chamber,

the first gas is pumped to the second reaction chamber, the first gas is ignited by the ignitor in the second reaction chamber to generate the propulsion gas.

11. The method of claim 9, wherein the thermal engine system further comprises a reaction chamber comprising a heat exchanger; and a boiler surrounding the reaction chamber, wherein the boiler contains water;

the reaction chamber generates heat from electrostatic fusion,

the heat exchanger transfers the heat from electrostatic fusion to the water in the boiler to vaporize the water into steam, and

wherein the steam is the propulsion gas.

12. The method of claim 9, wherein the thermal engine system further comprises a microturbine generator, wherein the microturbine generator generates power from the flow of the propulsion gas to the eductor.

13. The method of claim 9, wherein the propulsion gas is steam, carbon dioxide, or a combination thereof.

14. The method of claim 9, wherein the eductor provides control of thrust.

15. The method of claim 9, wherein the underwater propulsion thermal engine system comprises a plurality of eductors, wherein the plurality of eductors provides steering, thrust, and a combination thereof.

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